

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate specializing in scalable statistical inference, probabilistic modeling, and statistical diagnostics for stochastically dependent data. Develops new methods for large, noisy, partially observed datasets and builds reproducible Python/PyTorch research systems, with C++/pybind11 integration for performance-critical computation.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected 05/2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Research Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing and restartable research pipelines

Quantitative Methods: Gaussian processes, probability and stochastic processes, scalable likelihood approximation, hypothesis testing, Monte Carlo simulation, MLE/GLS, numerical optimization, spatiotemporal modeling, spectral and eigenvalue diagnostics, regression and machine learning, model validation, and uncertainty quantification

RESEARCH EXPERIENCE

Ph.D. Dissertation Research - Scalable Statistical Inference and Diagnostics

Sep 2024 - Present

- **Scalable Inference:** Developed an advection-aware Vecchia likelihood approximation for nonseparable space-time covariance structures, using ordered local conditional models to support likelihood-based parameter estimation and uncertainty quantification for up to 145,008 observations per day.
- **Stochastic Diagnostics:** Developed statistical diagnostic tools for dependent stochastic processes that localize where a fitted model fails across scale and frequency, distinguishing persistent dependence from high-frequency noise, missingness, acquisition artifacts, and covariance misspecification.
- **Research Systems:** Built end-to-end Python/PyTorch workflows for quality filtering, missingness tracking, time-dependent coordinate alignment, regular-grid matching, optimization, simulation, and restartable CPU/GPU and HPC execution; integrated performance-critical C++ ordering routines through pybind11.

QUANTITATIVE INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed a sequential risk-monitoring framework for binary indicators, using Brownian-motion approximation, dynamic-programming probability propagation, and simulation to quantify stopping-time uncertainty and early-decision tradeoffs.
- Specified and validated a Sequential Probability Ratio Test with explicit hypotheses and likelihood-ratio boundaries; evaluated boundary-crossing probabilities, stopping behavior, and Type I/II error.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2; presented findings as pricing and risk-segmentation recommendations to business stakeholders.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)